Exoplanet Hunter

The Search of New Home

Greeshma
General Engineering
IIT Mandi
Telangana, India
b26421@students.iitmandi.ac.in

Saanvi Tambde
General Engineering
IIT Mandi
Maharashtra, India
b26434@students.iitmandi.ac.in

Abhishek Birajdar
Electrical Engineering
IIT Mandi
Maharashtra, India
B26318@students.iitmandi.ac.in

Neha Patil
Engineering Physics
IIT Mandi
Maharashtra, India
b26402@students.iitmandi.ac.in

INTRODUCTION

By using provided data related to the three stars we have tried and concluded by reviewing all the aspects that are necessary to have a exoplanet around any star. Our project includes all the necessary calculations and statements to prove our conclusion. It is very necessary to find such a exoplanet because who knows, now leia needs new planet for survival of her species humans might need it someday…..

I. CONCLUSION FROM THE OBSERVATIONS

Based on the observations, STAC-2 and STAC-3 can be considered possible exoplanet candidates because their brightness dips occur at approximately regular intervals. STAC-2 shows dips at days 20, 100, and 180, indicating a period of about 80 days. STAC-3 shows dips near days 20, 60, 100, and 140, suggesting a period of about 40 days; the dip recorded at day 170 does not fit this pattern exactly and may reflect other stellar activity or an irregular measurement. These patterns support further investigation of both systems. Here we are assuming the orbital period of exoplanet of stac 3 to be 40 days

Based on the observations so far, the approximate orbital periods are 80 days for STAC-2 and 40 days for STAC-3.

Determination of Transit Depth

Transit depth represents the decrease in a star's relative brightness when a possible planet passes in front of it. It can be calculated using:

$$\delta = (F_{\text{normal}} - F_{\text{transit}}) \ / \ F_{\text{normal}}$$

For STAC-2

Normal relative brightness = 1.0000; relative brightness during transit = 0.9990.

$$\delta = (1.0000 - 0.9990) \ / \ 1.0000 = 0.0010 = 0.10\%$$

Therefore, the transit depth of STAC-2 is approximately 0.0010, or 0.10%.

For STAC-3

The recorded dips range from 0.9990 to 0.9993. Using a transit brightness of approximately 0.9991 and normal brightness of 1.0000:

$$\delta = (1.0000 - 0.9991) \ / \ 1.0000 = 0.0009 = 0.09\%$$

Therefore, the estimated transit depth of STAC-3 is approximately 0.0009, or 0.09%.

Reason for the Periodic Decrease in Stellar Brightness

When a planet passes in front of its star, it blocks some of the star's light and the observed brightness decreases slightly. Repeated decreases at approximately regular intervals can indicate a transiting planet. STAC-2 has a clearer 80-day pattern, while STAC-3 suggests a roughly 40-day pattern with one inconsistent observation, so both remain candidates rather than confirmed planets.

II. DETERMINING THE PLANET SIZE

From the observations, STAC-2 and STAC-3 are possible exoplanet candidates. We estimate the radii of the possible planets around these two stars using their transit depths.

The formula used is:

$$\delta = (\text{Rp} \ / \ \text{Rs})^{\,2}$$

Here, Rp is the radius of the planet, Rs is the radius of the star, and δ is the transit depth. Rearranging gives:

$$\text{Rp} = \text{Rs} \sqrt{\delta}$$

For STAC-2

The radius of STAC-2 is 0.80 R☉, and its transit depth is 0.0010.

$$\text{Rp} = 0.80\ \text{R}\odot \sqrt{0.0010} = 0.02530\ \text{R}\odot$$

$$\text{Rp} = 0.02530 \times 109.1\ \text{R}\oplus = 2.76\ \text{R}\oplus = 17{,}600\ \text{km}$$

Therefore, the estimated radius is about 2.76 Earth radii (approximately 17,600 km). Based on radius alone, it may be classified as **super-Earth-sized**.

For STAC-3

The radius of STAC-3 is 0.50 R☉, and its estimated transit depth is 0.0009.

$$\text{Rp} = 0.50\ \text{R}\odot \sqrt{0.0009} = 0.0150\ \text{R}\odot$$

$$\text{Rp} = 0.0150 \times 696{,}000\ \text{km} = 10{,}440\ \text{km} = 1.64\ \text{R}\oplus$$

Therefore, the estimated radius is about 10,440 km, or 1.64 Earth radii. Based on radius alone, it may be classified **as super-Earth- sized**

III. DETERMINING THE PLANETARY ORBITS

We have identified STAC-2 and STAC-3 as possible exoplanet candidates. We now estimate their orbital distances using Kepler's Third Law.

The simplified formula derived from Kepler's Third Law is:

$$P^2 = A^3 / M$$

Here, P is the orbital period in years, A is the orbital distance in astronomical units (AU), and M is the stellar mass in solar masses which gives us

$$A = \sqrt[3]{(MP^2)}$$

For STAC-2

Orbital period = 80 days; stellar mass = 0.80 M☉.

$$P = 4 / 365.25 = 0.0109514 \text{ years}$$

$$A = \sqrt[3]{[1.00 \times (0.0109514)^2]} = 0.0493 \text{ AU}$$

$$A = 0.0493 \times 149.6 \text{ million km} = 7.38 \text{ million km}$$

Therefore, **the orbital distance is 0.337 AU, or 50.4 million km.**

For STAC-3

Orbital period = 40 days; stellar mass = 0.50 M☉.

$$P = 40 / 365.25 = 0.1095 \text{ years}$$

$$A = \sqrt[3]{[0.50 \times (0.1095)^2]} = 0.1817 \text{ AU}$$

$$A = 0.1817 \times 149.6 \text{ million km} = 27.18 \text{ million km}$$

Therefore, the estimated **orbital distance is approximately 0.1817 AU, or 27.18 million km**.

The orbital distance of the possible planet around STAC-2 is approximately 3.0 times smaller than Earth's orbital distance from the Sun.

The estimated orbital distance of the possible planet around STAC-3 is approximately 5.5 times smaller than Earth's orbital distance from the Sun.

Determining Stellar Energy Received Relative to Earth

The stellar energy received by a planet relative to Earth is calculated using:

$$E_p / E_\oplus = (L / L\odot) / A^2$$

Here, L is the host star luminosity in solar luminosities and A is the orbital distance in AU. Earth receives 1 unit of stellar energy at 1 AU.

For STAC-2

Stellar luminosity = 0.40 L☉; orbital distance = 0.337 AU.

$E_p / E_\oplus = 0.40 / (0.337)^2 = 0.40 / 0.113569 = 3.52$

Therefore, the possible exoplanet around STAC-2 receives 3.52 times the stellar energy received by Earth.

For STAC-3

Stellar luminosity = 0.08 L☉; orbital distance = 0.1817 AU (based on the provisional 40-day period).

$E_p / E_\oplus = 0.08 / (0.1817)^2 = 0.08 / 0.03301489 = 2.42$

Therefore, assuming the provisional orbital distance**, the possible exoplanet around STAC-3 receives 2.42 times the stellar energy received by Earth.**

IV. MOST PROMISING EXOPLANET

STAC-2 has an estimated planet radius of 2.76 Earth radii and an orbital distance of approximately 0.337 AU. Its estimated orbital period is 80 days. The host star has a luminosity of 0.40 L☉, so the possible planet receives approximately 3.52 times the stellar energy received by Earth.

The possible planet around STAC-2 orbits slightly closer to its star than Mercury orbits the Sun (Mercury's orbital distance is about 0.387 AU). Its relatively high estimated stellar energy input may make Earth-like surface conditions challenging, although the actual temperature depends on its atmosphere, reflectivity, and heat distribution.

STAC-3 has an estimated planet radius of approximately 1.64 Earth radii and an estimated orbital distance of 0.1817 AU. If its provisional orbital period is about 40 days, the planet receives approximately 2.42 times the stellar energy received by Earth, using the supplied stellar luminosity of 0.08 L☉. This is lower than the estimated energy received by the possible planet around STAC-2.

Based on the simplified habitable-zone comparison and the lower estimated stellar energy input, STAC-3 may be the more promising candidate for further habitability studies. Neither candidate can be confirmed as habitable without additional observations of its atmosphere, composition, and surface conditions.

TABLE I

COMPARISON OF CANDIDATE EXOPLANETS

Property	STAC-2 Exoplanet	STAC-3 Exoplanet	Earth	Mercury
Host star mass	0.80 M☉	0.50 M☉	1.00 M☉	1.00 M☉
Host star radius	0.80 R☉	0.50 R☉	1.00 R☉	1.00 R☉
Host star luminosity	0.40 L☉	0.08 L☉	1.00 L☉	1.00 L☉
Host star surface temperature	5000 K	3800 K	5778 K	5778 K
Transit depth	0.10%	0.09%		
Planet radius	2.76 R⊕	1.64 R⊕	1.00 R⊕	0.383 R⊕
Orbital period	80 days	40 days	365.25 days	88 days
Orbital distance	0.337 AU	0.1817 AU	1.00 AU	0.387 AU
Orbital distance (million	50.4	27.18	149.6	57.9

km)				
Stellar energy received relative to Earth	3.52 times	2.42 times	1	6.67 times
Average surface temperature	Unknown	Unknown	15 °C	167 °C*

Notation: R⊕ = Earth radius; R⊙ = solar radius; M⊙ = solar mass; L⊙ = solar luminosity; AU = astronomical unit.

Note: The surface temperatures of the possible planets around STAC-2 and STAC-3 are unknown. Mercury's average surface temperature is approximately 167 °C, but it varies greatly between day and night. The stellar energy values for the candidates are estimates based on the supplied luminosities and calculated orbital distances; STAC-3's value depends on its provisional 40-day period.

The possible planet around STAC-2 has an orbital distance of approximately 0.337 AU and an orbital period of 80 days, relatively close to Mercury's 0.387 AU and 88-day orbit. However, it receives approximately 3.52 times Earth's stellar energy. The possible planet around STAC-3 is estimated to orbit at 0.1817 AU with a provisional period of about 40 days and receives approximately 2.42 times Earth's stellar energy. These estimates suggest lower stellar energy input for STAC-3, though its period requires confirmation.

The estimated radius of the possible planet around STAC-2 is 2.76 Earth radii, while that of STAC-3 is approximately 1.64 Earth radii. STAC-3's smaller estimated radius and lower stellar energy input make it an interesting candidate for further investigation, but the uncertainty in its transit timing should be considered.

Considering these factors together, STAC-3 appears to be the more promising candidate for further habitability studies based on the current simplified estimates. This conclusion is tentative because the day-170 brightness dip does not align with the otherwise approximately 40-day sequence. More observations are needed before its orbital period or habitability can be assessed confidently.

Additional Measurement for Assessing Habitability

The Habitable Zone (HZ) is the region around a star where a planet could potentially have liquid water on its surface, provided that suitable atmospheric conditions exist. A simplified estimate of the habitable-zone reference distance can be calculated using the star's luminosity:

$$D_{HZ} = \sqrt{(L^* / L\odot)}\ AU$$

Here, d_HZ is the approximate reference distance, L* is the luminosity of the star, and L⊙ is the luminosity of the Sun.

For STAC-2

The luminosity of STAC-2 is 0.40 L⊙.

$$D_{HZ} = \sqrt{(0.40\ L\odot / L\odot)} = 0.632\ AU$$

The estimated orbital distance of its possible planet is 0.337 AU. This is closer to its star than the simplified reference distance of 0.632 AU.

For STAC-3

The luminosity of STAC-3 is 0.08 L⊙.

$$D_{HZ} = \sqrt{(0.08\ L\odot / L\odot)} = 0.283\ AU$$

The estimated orbital distance of its possible planet is approximately 0.1817 AU. This is closer to its star than the simplified reference distance of 0.283 AU, but proportionally closer to the reference distance than STAC-2 is to its own.

TABLE II
HABITABLE-ZONE REFERENCE DISTANCE

Property	STAC-2	STAC-3
Stellar luminosity	0.40 L⊙	0.08 L⊙
Estimated planetary orbital distance	0.337 AU	0.1817 AU
Simplified HZ reference distance	0.632 AU	0.283 AU
Position relative to reference distance	Closer to star	Closer to star but at the same time nearer to reference distance

Conclusion

According to this simplified comparison, neither candidate orbits at the reference distance where it would receive approximately the same stellar energy that Earth receives from the Sun. STAC-2 orbits at 0.337 AU compared our calculated habitable zone distance of 0.632 AU, while STAC-3 is estimated to orbit at 0.1817 AU compared with a reference distance of 0.283 AU.

Therefore, STAC-3 is the more promising candidate for further habitability studies based on the comparisons and also it absorb less stellar energy. But their can be question related to it orbital distance from stac 3 and related to its orbital period . But it's size and amount of stellar energy it absorb can counter balance these to factor and we can assume that it won't lead to increase in it's surface temperature to large extent. But to find or to be firmly apply it we need more information about it's surface and temperature. And also star stac 3 is appear to be in it's prime phase as we can see from it's observational data since it's size is smaller than our sun it will use it's hydrogen at slower rate as compared to sun.